Geographical Data and Computational Tools for Determining Restoration Priorities in Mangrove Ecosystems: A Scoping Review

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Abstract

Mangrove ecosystems provide critical ecosystem services, including carbon sequestration, coastal protection, and host high level of biodiversity, yet they continue to experience widespread degradation. Advances in Earth observation and computational analysis have enabled increasingly detailed monitoring of mangrove extent, structure, and dynamics using multi-source geospatial data, alongside the growing adoption of machine learning methods. This scoping review synthesises the current landscape of geographical data and computational tools used to assess mangrove conservation status and restoration needs. We reviewed peer-reviewed studies applying geospatial data in combination with machine learning techniques and charted them by data sources, sensing modalities, analytical methods, evaluation practices, and application scope. While substantial progress has been made in data availability and modelling capability, the literature remains dominated by region-specific studies, with explicit cross-region evaluation or transfer learning rarely reported. Most analysis rely on individual sensing modalities, highlighting the absence of unified multimodal frameworks. Additional limitations include reduced robustness in fragmented landscapes, limited characterisation of submerged or intertidal zones, and minimal incorporation of in situ validation and local ecological knowledge. These gaps constrain the generalisability and restoration relevance of current approaches, underscoring the need for transferable, multimodal, and decision-oriented computational frameworks for mangrove ecosystem assessment and restoration prioritisation.

Introduction

Mangrove forests are among the most productive and valuable coastal ecosystems, performing essential ecological functions such as shoreline stabilisation, storm surge attenuation, carbon sequestration, nutrient cycling, and providing habitat for highly diverse biological communities [1].

Mangroves also serve as natural coastal defence barriers, providing protection against tidal erosion and storm surges while functioning as habitats for numerous species of fish and birds. In doing so, they support coastal communities worldwide and contribute significantly to global fisheries [1].

Despite their importance, as highlighted in *The State of the World's Mangroves 2024*, these ecosystems continue to experience widespread degradation driven by aquaculture expansion, coastal development, altered hydrological regimes, pollution, and climate-induced stressors, including sea-level rise and increasing temperature extremes [1].

Over the past two decades, substantial progress has been made in monitoring mangrove ecosystems through advances in Earth observation technologies and computational analysis [2]. Global and regional datasets derived from optical satellite imagery, synthetic aperture radar (SAR), LiDAR, hyperspectral sensors, and unmanned aerial vehicles have enabled increasingly detailed assessments of mangrove extent, structure, and temporal dynamics.

In parallel, machine learning and deep learning methods have been widely adopted to improve classification accuracy, detect change, and estimate biophysical properties such as canopy height and biomass [3], [4], [5], [6], [7], [8], [9], [2], [10], [11], [12], [13], [14], [15], [16] [17], [18], [19].

Collectively, this body of work has significantly advanced the ability to observe mangrove forests across multiple spatial and temporal scales.

However, these advances have largely evolved in isolation. Existing studies typically focus on specific sensing modalities, geographic regions, or analytical tasks, with limited integration across data sources or modelling approaches. As a result, much of the literature remains oriented toward mapping and monitoring, rather than toward decision relevant assessment of restoration needs. In particular, the growing diversity of datasets and computational tools has not been systematically synthesised to evaluate how effectively current approaches support restoration prioritisation.

To address these challenges, this scoping review systematically synthesises existing research on mangrove ecosystem monitoring and the computational analysis of the resulting data. Specifically, the objec-

tives of this review are to:

- Examine the completeness and robustness of existing geospatial datasets used for mangrove analysis.
- Identify the sensing technologies employed to acquire mangrove data, including optical, SAR, LiDAR, hyperspectral, and thermal modalities.
- Review the computational and machine learning techniques applied to these datasets, along with their reported performance metrics.
- Outline current limitations associated with data acquisition and computational analysis approaches.
- Identify key research gaps related to regional bias, data availability, and methodological limitations in current mangrove research.

A scoping review approach is appropriate for this study because the literature on mangrove restoration spans diverse datasets, sensing technologies, and computational methods, with heterogeneous objectives, study designs, and evaluation metrics that preclude narrow systematic synthesis. Accordingly, this paper aims to identify methodological trends and highlight research gaps within the current ecological and computational frameworks applied to mangrove ecosystems.

This review does not propose or evaluate a specific computational model. Instead, it synthesises existing evidence to identify limitations in current practice and to outline future research directions for mangrove ecosystem analysis.

As discussed in subsequent sections, several recurring methodological and data-related limitations emerge across the reviewed literature, which collectively constrain the scalability and restoration relevance of current mangrove ecosystem assessment approaches.

Methodology

This study adopts a scoping review methodology following the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses - Scoping Reviews) guidelines [20]. A scoping review approach was selected due to the interdisciplinary nature of the research topic, which spans geospatial data acquisition, remote sensing technologies, and computational analysis techniques applied to mangrove ecosystems. This approach enables comprehensive mapping of existing research, identification of methodological trends, and systematic detection of research gaps without restricting inclusion to narrowly defined outcome measures.

Literature Search Strategy

A structured literature search was conducted across five academic databases: *SpringerLink*, *IEEE Xplore*, *EBSCO*, *Scopus*, and *Nature*. These databases were selected to ensure broad coverage of environmental science, Earth observation, and machine learning literature.

Search queries were designed to capture studies related to mangrove ecosystems, geospatial data acquisition, and computational modelling. Keywords included combinations of terms such as *mangrove*, *remote sensing*, *satellite*, *SAR*, *LiDAR*, *hyperspectral*, *thermal*, *machine learning*, *deep learning*, and *convolutional neural networks*. The search strategy was intentionally broad to maximise recall, consistent with scoping review best practices.

Study Selection and Screening

All records retrieved from the database searches were aggregated and screened in multiple stages. Duplicate records were removed prior to screening. Non-English publications were excluded at this stage to ensure consistency in analysis.

The remaining records underwent title and abstract screening. Because the authors's institution provides full-text access to all searched databases, no studies were excluded due to retrieval limitations. The screened studies were then divided among members of the research team, and each reviewer independently assessed titles and abstracts for relevance to the research objectives.

Studies were excluded during screening if they:

- Did not involve geospatial or remotely sensed data.
- Relied solely on descriptive or basic statistical analyses without computational modelling.
- Did not employ modelling approaches central to this review, such as machine learning or deep learning methods (E.G., Random Forest, convolutional neural networks).

Disagreements between reviewers were resolved through discussion to ensure consistency in study selection.

Eligibility Assessment and Inclusion

Following the screening stage, the remaining studies were assessed at the full-text level for eligibility. Studies were excluded if they lacked a clear spatial component, did not contribute substantively to understanding mangrove ecosystem monitoring or restoration, or fell outside the scope of computational and geospatial analysis considered in this review.

After applying the eligibility criteria, a final set of studies was retained for inclusion in the scoping review. These studies form the basis for the qualitative synthesis and quantitative analysis presented in subsequent sections.

Article Selection Process

The overall study selection process is summarised in the PRISMA flow diagram shown in Figure 1. The diagram illustrates the identification of records across databases, removal of duplicates, screening of titles and abstracts, assessment of full-text eligibility, and final inclusion of studies in the review.

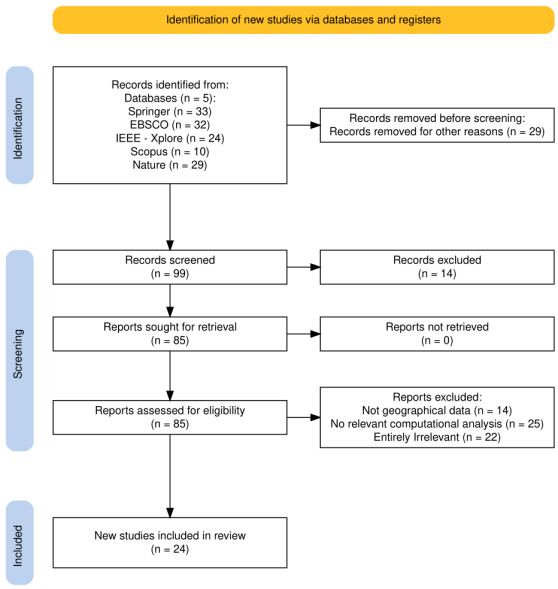

Figure 1: *PRISMA flow diagram illustrating the identification, screening, eligibility assessment, and inclusion of studies in this scoping review.*

Note

In addition to the studies included through the formal screening process, a limited number of supplementary references were used to provide background context on established datasets, sensing modalities, and foundational concepts. These sources were not analysed as part of the scoping review evidence base and were included solely to support clarity, terminology, and methodological description.

Current Data Availability and Dataset Robustness

Evolution of Global Mangrove Datasets

Global mangrove datasets underpin large scale conservation assessments and policy initiatives. Early global products, such as the World Mangrove Atlas and the Landsat-based map [21], provided the first consistent estimates of global mangrove extent. While these datasets established critical baselines, they were static snapshots and did not capture temporal dynamics.

The introduction of time-series datasets marked a significant methodological shift. The Continuous Global Mangrove Forest Cover for the 21st Century (CGMFC-21) provided annual mangrove extent maps from 2000 to 2014 at 30 m resolution, enabling year-by-year analysis of loss and gain [8]. However, CGMFC-21 employed conservative mangrove definitions, resulting in substantially lower area estimates than other products and potentially underrepresenting sparse or degraded stands.

The Global Mangrove Watch (GMW) initiative represents the most comprehensive effort at the present time. It integrates optical and SAR data to produce multi-epoch and near-global time-series products. The latest release, GMW v4.0, established a 10 m resolution global baseline for 2020 with reported overall accuracy exceeding 95% [22]. This improvement significantly enhances the detection of narrow, fragmented, and small mangrove patches. Table 1 provides a comparative overview of major global mangrove datasets, summarising their release timelines, spatial resolution, temporal coverage, primary data sources, and the ecosystem attributes they capture. As per the latest report issued by the Global Mangrove Watch, this dataset retains its standing among the conservation community [1].

Dataset Robustness and Uncertainty

Despite methodological advances, global mangrove datasets remain subject to several limitations. Differences in sensor selection, classification algorithms, training data, and mangrove definitions lead to discrepancies in mapped extent. This is seen particularly near latitudinal range limits and in transitional ecosystems [1]. Temporal gaps between dataset updates may obscure short-term disturbances or rapid regeneration events, while most global products prioritise extent over structural or functional attributes. This last point is an important observation, as it is ofter times that functional attributes that translate into ecological services (fishing grounds, water filtration, resources, etc.) are what truly moves communities and authorities to act pro or against conservation efforts. Assessing these attributes is thus of great relevance.

Lastly, validation data is unevenly distributed geographically. Regions with limited field data often exhibit higher uncertainty, which can propagate into downstream analyses. This can only be addressed

Table 1: *Major global mangrove datasets and their key characteristics*

Dataset	First Release	Spatial Resolution	Temporal Coverage	Primary Data Sources	Mangrove Attributes Captured
Global Mangrove Forests Distribution	2000	∼30 m	Circa 2000	Landsat	Extent
CGMFC-21 [8]	2014	∼30 m	2000–2012	Landsat-derived products	Annual extent and change
Global Mangrove Watch v2.0 [23]	2019	20–25 m	1996–2016	ALOS PALSAR, Landsat	Extent and loss
Global Mangrove Watch v3.0 [22]	2022	∼25 m	1996–2020	Multi-mission SAR and optical	Extent and temporal dynamics
Global Mangrove Canopy Height Maps [24]	2020+	∼30 m	Multi-year	DEM and optical integration	Canopy height and structure
GEDI [25]	2019	∼25 m LiDAR footprints	2019–2023	Spaceborne LiDAR (GEDI)	Canopy height, vertical structure, biomass proxies
Global Mangrove Soil Carbon Stocks [26]	2023	∼30 m	2020	Remote sensing and modelling	Soil organic carbon

Note: GEDI provides discrete LiDAR footprints rather than continuous spatial coverage and is therefore typically integrated with wall-to-wall mangrove extent datasets for analysis.

with field work and completion of datasets. As stated by the Mangrove Global Watch, in locations where data fails, it is crucial to rely on local communities and traditional knowledge [1]. This provides validation (or points out inaccuracies in the models), and makes the scientific efforts tangible to society.

Geological Survey Tools and Data Acquisition Technologies

Mangrove ecosystems are monitored using a diverse suite of geospatial sensing technologies, each capturing distinct and complementary attributes of forest condition (physiological stress, drought, etc,) or structure (plant height and density, topological features, etc.)[3], [1]. No single sensing modality among the retrieved papers provides a complete description of mangrove ecosystems, making multimodal integration a notable hole in the field, and a milestone for conservation science.

Thus, a holistic representation would refer to a joint latent encoding that simultaneously captures forest extent, vertical structure, physiological condition, hydrological dynamics, and temporal change, enabling restoration prioritisation based on both current state and predicted developments.

Figure 2 provides an overview of the major sensing modalities used in mangrove studies and the ecological attributes they capture. The matrix highlights that individual sensing modalities provide partial and complementary views of mangroves.

Sensing modalities	Ecological attributes						
	Areal extent and spatial distribution	Canopy structure and height	Biomass and carbon stocks	Species composition and biochemical traits	Hydrology and inundation dynamics	Physiological stress and temperature anomalies	Temporal change and disturbance recovery
Optical multispectral imagery	●●	●	●	●	○	○	●●
Synthetic Aperture Radar (SAR)	●	●●	●	○	●●	○	●●
LiDAR	●	●●●	●●	○	○	○	●
Hyperspectral imagery	●	●	●	●●●	○	●	●
Thermal infrared	○	○	○	○	●	●●●	●
UAV-based sensing	●●	●●	●●	●	●	●	●●

Legend:
●●● = strong information ●● = moderate information ● = limited information ○ = minimal or indirect information

Figure 2: *Conceptual modality attribute matrix illustrating the relative strength of ecological information captured by different sensing modalities in mangrove ecosystem studies[21][27][28][29][30][31][32][33][34].*

Geospatial and Remote Sensing Tools Used in Mangrove Research

Table 2 summarises the aggregated frequency of major geospatial and sensing tools used in mangrove studies across the surveyed databases.

Several clear trends emerge from Table 2. First, satellite-based and airborne optical and radar systems dominate mangrove research. High-resolution aerial imagery and SAR-based platforms (particularly ALOS-PALSAR) appear most frequently. This is likely due to the need to assess large geographical spaces and relying on inundation dynamics to classify mangrove ecosystems.

Noticeably, advanced sensing modalities such as

Tool / Data Source	Number of Studies
High-Resolution Aerial Imagery	2431
ALOS-PALSAR (SAR)	2205
Thermal SAR (T-SAR)	1799
Thermal Infrared Sensors (TIRS)	1120
Hyperspectral Imaging (HSI)	812
SAR (general)	449
Satellite Platforms (general)	266
CubeSats	174
LiDAR	123
Landsat	28

Table 2: *Summary of tools and data sources used in mangrove studies*

LiDAR and hyperspectral imaging remain comparatively underrepresented, despite their demonstrated ability to capture three-dimensional structure, species composition, and biochemical traits [11], [35]. This suggests that while high-fidelity structural and functional data are recognised as valuable, their integration into mainstream mangrove monitoring workflows remains limited, likely due to cost, availability, and analytical complexity.

Third, the relatively low representation of Landsat compared to SAR-based and aerial platforms reflects a gradual shift away from coarse optical imagery toward sensors better suited to cloud-prone dynamic coastal environments. Overall, these patterns indicate a strong dependence on a narrow subset of sensing technologies, which further states the need for multimodal integration to capture the full complexity of mangrove ecosystems.

Optical Multispectral Remote Sensing

Optical multispectral imagery from platforms such as Landsat and Sentinel-2 remains the most widely used data source for mangrove monitoring. These sensors support vegetation indices, phenological analysis, and land-cover classification [22]. Sentinel-2's red-edge bands enhance discrimination between mangrove vegetation and adjacent land covers but remain susceptible to cloud cover and atmospheric interference [19].

Synthetic Aperture Radar (SAR)

Synthetic Aperture Radar (SAR) is an active remote sensing technology operating in the microwave domain, enabling cloud-independent observation and consistent data acquisition in tropical regions. SAR backscatter is sensitive to vegetation structure, moisture content, and inundation dynamics, making it well suited for mangrove delineation and monitoring in persistently cloudy coastal environments. Sensors such as Sentinel-1 and ALOS PALSAR have demonstrated strong performance in detecting narrow and fragmented mangrove stands and in capturing tidal and hydrological variability, which is particularly relevant for restoration suitability assessment [22].

LiDAR

LiDAR systems enable direct measurement of canopy height, vertical stratification, and biomass. Li demonstrated that integrating LiDAR with multispectral and hyperspectral data substantially improves multilayer mangrove structure mapping [11]. Despite its accuracy, LiDAR remains spatially sparse and costly, limiting its operational use at national or global scales.

Hyperspectral and Thermal Infrared Sensors

Hyperspectral data support species-level discrimination and biochemical trait estimation, while thermal infrared data reveal physiological stress and hydrological anomalies [33], [35]. These modalities are underutilised in large-scale mangrove studies but offer significant potential for early stress detection and restoration monitoring.

UAV-Based Observations

UAV platforms provide ultra-high spatial resolution imagery for local-scale assessment and validation. While invaluable for monitoring restoration sites, UAV data lack scalability and require integration with satellite-based observations [36].

Computational Analysis Techniques

Machine learning and deep learning methods are increasingly used to analyse mangrove-related geospatial datasets for both regression and classification tasks, driven by the growing availability of multi-source remote sensing data and the need to infer biophysical properties at scale [37][21], [8]. Regression tasks typically aim to estimate continuous variables such as above-ground biomass, carbon stocks, or canopy height, whereas classification tasks focus on mangrove extent delineation, species discrimination, and change detection [30][29]. Across these applications, ensemble-based methods, kernel-based learners, and deep neural networks have emerged as dominant computational approaches.

Regression-based modelling

Regression-based machine learning models are widely applied to estimate mangrove biomass and structural attributes from remotely sensed inputs, with Random Forest, Support Vector Machine, and gradient boosting methods among the most commonly used approaches [13], [38], [39]. These models are well suited to capturing non-linear relationships between spectral, textural, and structural features [40] [41][42]. Model performance in such studies is typically evaluated using the coefficient of determination (R^2) and root mean square error (RMSE)[11], [13].

Empirical results reported in the mangrove literature indicate moderate to strong predictive performance for biomass estimation, though with substantial variability across regions and forest types. For example, Maung [13] evaluated RF, SVM, and XGBoost models for above-ground biomass estimation in Myanmar mangroves using multispectral and SAR inputs and reported that RF achieved the best performance (R^2 = 0.48, RMSE = 28.12 Mg ha^{-1}) after feature selection, while alternative models exhibited markedly lower explanatory power. In a contrasting subtropical plantation setting, Peng reported higher accuracy using XGBoost applied to Sentinel-2 data, achieving an R^2 of approximately 0.68 with an RMSE of 6.85 Mg ha^{-1}, reflecting the reduced structural complexity and biomass range of managed stands[43]. In structurally complex old-growth mangroves, Selvaraj reported an R^2 of 0.78 but a substantially higher RMSE of 38.24 Mg ha^{-1}, highlighting the influence of biomass heterogeneity and signal saturation effects [44].

Across studies, reported R^2 values for mangrove biomass estimation commonly fall within the range of 0.5-0.8, with RMSE values strongly dependent on forest maturity, biomass distribution, and sensor configuration [39] [18]. Multiple studies demonstrate that regression performance improves when complementary structural information is incorporated. In particular, the integration of canopy height metrics derived from airborne or spaceborne LiDAR with optical and SAR features has been shown to mitigate saturation effects and improve predictive accuracy [45] [39] [18]. These findings consistently support the value of multimodal data fusion for robust estimation of mangrove biophysical properties.

Classification-based modelling

Classification approaches are extensively used for mangrove extent mapping, species classification, and land cover change detection. Traditional machine learning models such as Random Forest and SVM remain widely adopted due to their robustness to noisy inputs and limited training data requirements [46] [21]. More recently, convolutional neural networks (CNNs) and other deep learning architectures have been applied for pixel-level segmentation and fine-scale species mapping, particularly where high-resolution optical or hyperspectral imagery is available [2], [11].

Reported classification performance is typically quantified using overall accuracy, Cohen's Kappa, F1-score, or area under the receiver operating characteristic curve (AUC), depending on task formulation [47]. Representative studies indicate that high classification accuracy is achievable under favourable conditions. For instance, Aparicio demonstrated strong performance of a lightweight CNN architecture (MobileNetV3) for mangrove species identification from smartphone imagery, illustrating the potential of deep learning for in situ and participatory monitoring applications [3]. In remote sensing–based studies, Random Forest classifiers frequently achieve overall accuracies exceeding 90% when trained on high-resolution multispectral or hyperspectral data, while SVM performance is more sensitive to kernel selection and training sample composition [37] [35].

Across the literature, overall classification accuracies for mangrove mapping commonly range between 80% and 95%, with higher values often reported in studies covering limited spatial extents or a small number of species classes [2], [35]. However, several studies caution that high overall accuracy can obscure systematic misclassification of rare species or fragmented mangrove patches, emphasising the importance of reporting class-level metrics and confusion matrices [47] [46].

Evaluation practices and limitations

Despite advances in computational modelling, evaluation practices across mangrove studies remain heterogeneous. Most studies rely on random cross-validation or hold-out test sets drawn from the same geographic region used for model training, while cross-region or cross-site validation is rarely conducted [13], [35]. As a result, reported performance metrics may overestimate model generalisability, particularly in geographically heterogeneous mangrove systems where spectral, structural, and environmental conditions vary substantially [29] [39].

Differences in target variables, sensor combinations, spatial resolution, and validation protocols further limit direct comparison of reported metrics across studies [47]. Recent work has therefore highlighted the need for standardised benchmarking datasets, spatially explicit validation strategies, and transparent reporting of uncertainty [46][11]. En-

semble approaches and multimodal learning frameworks have been proposed as promising directions to improve robustness and reduce sensitivity to site-specific biases, although their application in mangrove research remains limited and largely region-specific [4], [11].

Research Gaps

Regional Bias

Most mangrove modelling studies train and evaluate models within a single geographic region. When applied to new regions, performance degradation is expected due to domain shift arising from differences in species composition, canopy structure, sediment characteristics, tidal regimes, and anthropogenic pressures. These factors induce systematic variation in optical reflectance, SAR backscatter, thermal signatures, and structural metrics, resulting in both covariate and concept shift across regions [11, 1].

Evidence from this scoping review indicates that explicit assessment of transferability remains rare. Across the five databases searched, only the studies listed in Table 3 were identified as primary research explicitly addressing transfer learning or cross-domain generalisation in the context of mangrove ecosystem analysis. Studies that discussed transferability only at a conceptual or review level were excluded from this count. As illustrated in Figure 3, which summarises evaluation practices across the included studies, the majority rely on single-region evaluation (23 studies), while a further 16 consider multiple regions without testing cross-region transfer. Only two studies explicitly investigate model behaviour beyond the training domain.

The FragMangro framework addresses weak cross-domain generalisation by proposing a zero-shot detection model applied across three ecologically distinct mangrove sites within the Guangxi Zhuang Autonomous Region. Although geographically co-located, the Beilun Estuary, Shankou, and Maowei Sea exhibit differing fragmentation levels, spatial configurations, and spectral mixing characteristics. By applying a single trained model across these sites without retraining, the study demonstrates robustness to domain shift driven by landscape fragmentation rather than geographic distance [19]. In contrast, Fu explores cross-region transfer learning for estimating mangrove leaf nitrogen and phosphorus content, showing that domain adaptation can reduce prediction error, but focusing narrowly on biochemical traits rather than ecosystem-scale properties relevant to restoration [35].

Taken together, these findings indicate that transferability has been addressed only in a small number of task-specific studies within the mangrove literature. The dominance of region-specific evaluation limits confidence in the scalability and restoration relevance of current approaches. Addressing regional bias therefore requires systematic cross-region benchmarking and the routine incorporation of transfer learning and domain-aware evaluation strategies, consistent with the broader need for generalisable, restoration-oriented modelling frameworks discussed in subsequent sections.

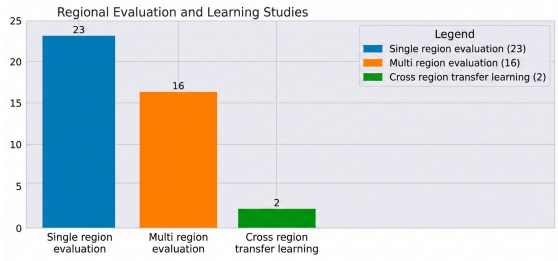

Figure 3: *Evaluation strategies used in the reviewed mangrove studies.*

Table 3: *Transferability across digital libraries*

Transferability	
Database	**No. of Papers**
SpringerLink	0
EBSCO	1
IEEE Xplore	1
Scopus	1

Lack of a Unified Multimodal Forest Framework

A second major research gap is the absence of a unified multimodal forest framework capable of integrating complementary geospatial data sources into a coherent and ecologically meaningful representation of mangrove ecosystems.

Each sensing modality captures a distinct and non-redundant component of forest condition:

- Optical multispectral imagery encodes extent, canopy cover, and phenology [22].
- SAR captures structure, moisture, and inundation dynamics under all-weather conditions [6].
- LiDAR provides direct measurements of three-dimensional canopy architecture and biomass [11].
- Thermal infrared data reveal physiological stress and hydrological anomalies [33].

- Hyperspectral data support biochemical and species-level trait estimation [35].

Single-modality analyses can therefore produce misleading restoration assessments. Intact canopy cover inferred from optical imagery may obscure physiological stress detectable in thermal data, while LiDAR-derived height alone does not capture hydrological constraints or salinity stress. Restoration decisions based on incomplete representations risk misidentifying resilient forests or prioritising unsuitable restoration sites.

Recent advances in multimodal representation learning offer a pathway toward addressing this limitation. Li et al. (2021) demonstrated that integrating multispectral, hyperspectral, and LiDAR data improves multilayer structural mapping, while Bahaduri et al. (2024) showed that channel-level cross-attention enables dynamic alignment between heterogeneous modalities. Channel-level fusion is particularly advantageous in mangrove ecosystems, where data relevance varies spatially and temporally due to cloud cover, tidal state, and sensor availability.

Figure 5 illustrates a conceptual unified multimodal framework for integrating spatiotemporal optical, SAR, LiDAR, thermal infrared, and multispectral or hyperspectral data into a shared latent forest representation, informed by the multimodal integration strategy proposed by Li et al. (2021).

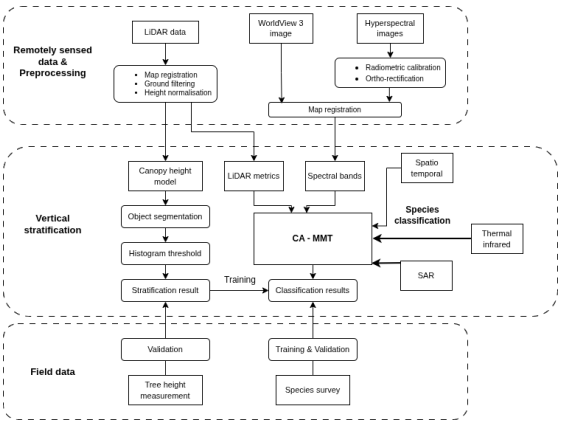

Figure 4: *Draft of what a unified multimodal framework for mangrove ecosystem surveying would look like [11].*

Importantly, this review does not advocate a single architectural solution. Rather, it highlights that a holistic representation of mangrove ecosystem condition cannot be obtained from any single sensing modality, as key attributes such as extent, structure, hydrology, physiological stress, and biochemical composition are captured across complementary optical, SAR, LiDAR, thermal infrared, and multispectral or hyperspectral data sources. Cross-attention transformers, multimodal convolutional networks, graph-based models, hybrid ConvLSTM-Transformer pipelines, and ensemble approaches all represent viable computational pathways depending on data availability and restoration objectives. The central gap lies not in the absence of individual techniques, but in the lack of end-to-end, restoration-oriented multimodal frameworks explicitly designed to fuse these complementary modalities into generalisable and decision-relevant mangrove ecosystem representations.

Limitations of CNN-Centric Approaches in Fragmented Mangrove Landscapes

Convolutional neural networks underpin most deep learning-based mangrove classification and segmentation pipelines. While effective for large, contiguous forest stands, CNN-based approaches often struggle to detect small, fragmented, or linear mangrove patches embedded within complex coastal environments. This limitation is particularly pronounced in regions experiencing urbanisation, aquaculture expansion, or partial degradation [19].

Fragmentation-related challenges arise from three interconnected factors: reduced spatial context at fixed receptive field sizes, spectral confusion with surrounding vegetation or built environments, and heavy reliance on dense manual annotations. These issues contribute to reduced robustness when models are transferred to regions with differing fragmentation patterns. Addressing this gap will require exploration of multi-scale architectures, weakly supervised or semi-supervised learning strategies, and alternative spatial representations that better capture fine-grained ecological structure [19].

Inadequate Observation of Submerged and Underwater Mangrove Components

Most satellite-based mangrove monitoring relies on visible and near-infrared wavelengths, which are strongly attenuated by water, limiting detection of submerged root structures and early-stage regeneration in intertidal and subtidal zones [48]. As a result, below-water components critical to ecosystem stability and recovery are often poorly characterised or entirely omitted from restoration assessments.

This sensing limitation constrains evaluation of mangrove health and restoration potential in tidally dynamic and turbid environments. Addressing this gap will require integration of alternative observation strategies, including UAV-based surveys, bathymetric and acoustic data, and improved fusion of radar and optical observations across tidal states [49],[29].

Limited Integration of Field Validation and Local Knowledge

Despite advances in geospatial modelling, field validation and local ecological knowledge remain insufficiently integrated into many mangrove restoration studies. In situ observations are essential for validating remotely sensed classifications, contextualising model outputs, and identifying site-specific constraints that may not be captured by remote sensing alone.

The absence of systematic field validation can result in restoration suitability maps that are technically accurate yet ecologically or socially misaligned. Future frameworks should therefore emphasise participatory validation, integration of field measurements, and collaboration with local stakeholders to ensure that computational outputs translate into context-aware and implementable restoration decisions [1].

Future Work: Toward Unified Multimodal Frameworks for Mangrove Restoration Prioritisation

Building on the research gaps identified in this review, future work should prioritise a shift from isolated, region-specific modelling pipelines toward unified, multimodal, and generalisable computational frameworks for mangrove ecosystem assessment. Addressing these gaps requires coordinated advances in data integration, model design, evaluation methodology, and translational practice.

A primary direction for future work is the development of *unified multimodal forest frameworks* that integrate complementary sensing modalities, including spatiotemporal optical imagery, SAR, LiDAR, thermal infrared, and multispectral or hyperspectral data. Because each modality captures distinct ecological attributes, multimodal integration enables holistic representations that cannot be achieved using any single data source. Multimodal integration enables the construction of holistic latent representations capable of supporting restoration prioritisation based on both current ecosystem condition and long-term resilience.

Recent advances in multimodal representation learning offer promising tools for achieving such integration. Attention-based architectures, including cross-attention multimodal transformers, allow heterogeneous data streams to interact at the feature or channel level, enabling dynamic weighting of modalities based on contextual relevance and data quality. This property is particularly valuable in mangrove environments, where sensor availability and informativeness vary spatially and temporally due to cloud cover, tidal state, and acquisition constraints. However, future research should remain architecture-agnostic, systematically comparing attention-based transformers, multimodal convolutional networks, graph-based models, and hybrid spatiotemporal pipelines under realistic data availability and domain shift conditions.

Addressing *regional bias and limited generalisation* represents another critical research direction. Future frameworks should explicitly incorporate cross-region validation, transfer learning, and domain adaptation strategies to assess and improve model robustness across diverse ecological contexts. Pretraining multimodal models on globally distributed datasets, followed by region-specific fine-tuning using limited local data, represents a promising pathway for reducing dependence on data-rich regions and improving scalability. Such strategies are critical for ensuring that restoration prioritisation models remain robust under domain shift and can be deployed beyond data-rich regions.

Future work must also move beyond CNN-centric pipelines that struggle in fragmented mangrove landscapes. Exploration of multi-scale representations, object-centric models, weakly supervised learning, and semi-supervised approaches may improve detection of small or degraded mangrove patches while reducing reliance on extensive manual labelling. Such approaches are particularly important in urbanised or partially restored coastal environments where fragmentation is pronounced.

The challenge of observing submerged and underwater mangrove components further motivates integration of alternative sensing strategies. Combining satellite observations with UAV-based surveys, bathymetric data, and tide-aware data fusion can improve characterisation of intertidal and subtidal zones. Developing models that explicitly account for tidal state and water-column effects will be essential for accurate assessment of early-stage regeneration and below-canopy structures.

Finally, future research should prioritise *restoration-oriented and participatory outputs*. Integrating field validation, in situ measurements, and local ecological knowledge into computational pipelines will be critical for ensuring that model outputs are actionable and context-aware. Restoration prioritisation frameworks should produce interpretable suitability indices, uncertainty estimates, and scenario-based projections that support decision-making by practitioners, policymakers, and local communities.

Conclusions

This scoping review synthesised the current state of geographical data, sensing technologies, and computational tools used to determine restoration priorities in mangrove ecosystems. By combining qualitative literature analysis with quantitative synthesis of tools and techniques across major academic databases, the review provides a comprehensive overview of methodological trends, strengths, and limitations in contemporary mangrove research.

The findings reveal that while satellite-based optical and SAR data dominate the literature and classical machine learning approaches remain prevalent, advanced sensing modalities such as LiDAR, hyperspectral, and thermal infrared data are comparatively underutilised. Similarly, despite growing interest in deep learning, most computational pipelines remain CNN-centric and single-modality, limiting their effectiveness in fragmented landscapes and complex ecological settings.

Five critical research gaps were identified. First, strong regional bias in data availability and model development constrains generalisation and scalability. Second, the absence of unified multimodal forest frameworks restricts the integration of complementary structural, functional, and temporal information. Third, CNN-based approaches struggle with small and fragmented mangrove patches and rely heavily on costly manual labelling. Fourth, current sensing strategies inadequately capture submerged and underwater mangrove components. Finally, limited integration of field validation and local ecological knowledge reduces the translational impact of computational outputs.

Addressing these gaps requires a shift toward unified, multimodal, and generalisable computational frameworks that integrate diverse sensing modalities, explicitly account for domain shift, and align model outputs with restoration decision-making needs. By framing mangrove restoration as a multimodal representation learning and generalisation challenge, this review positions future research at the intersection of remote sensing, ecology, and machine learning. Advancing in this direction is essential for translating technological progress into robust, scalable, and socially relevant strategies for mangrove conservation and restoration.

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Accompanying Report

Group 13

Challenges Faced and Solutions:

One major challenge we faced at the beginning of the project was our initial plan: we wanted to build a completely new predictive model from scratch. The idea was to identify which ecosystems and specific locations would give the highest conservation value if we could only reserve or manage a limited number of sites. In other words, under strong resource constraints, we wanted the model to tell us: “If you can only protect a few places, which ones matter the most?”

However, we soon realised that this approach was far more demanding than we had expected. To do it properly, the model would need to consider many interacting factors, including:

1. projected climate change and its impact on different regions,

2. current biodiversity patterns and ecosystem types,

3. levels of human disturbance or land-use pressure,

4. management and implementation costs, and

5. potential future risks such as extreme weather or habitat loss.

Developing a model that integrates all these variables needs careful methodological design, large high-resolution geospatial datasets and substantial computational resources for training and validation. In practice, we lacked access to the necessary data sources and sufficient computing power or time to execute such an intensive training process. These limitations in data availability, computing resources and project timescale became a significant challenge and compelled us to reconsider the scope and ambition of our original research plan.

After we abandoned the idea of building a new model and redefined the project, we decided to conduct a scoping review instead. The final topic of our review is:

“Current Status of Geographical Data and Computational Tools for Determining Restoration Priorities in Mangrove Ecosystems.”

At the very beginning of the scoping review, one of the main difficulties we faced was designing an effective search strategy. We were not just interested in studies using simple descriptive or basic statistical analyses. We specifically wanted to identify

papers that applied modelling approaches, such as Random Forest or CNN-based methods.

To address this, we started with relatively broad and simple queries and conducted a pilot search in a few databases. We briefly screened the titles and abstracts of the initial set of papers to see which methods were being used and how authors described their modelling approaches. Based on this quick reading, we identified common keywords related to geospatial modelling in mangrove ecosystems. We then refined our search strings by combining terms related to mangroves, geographical or remote-sensing data, and advanced computational methods. Through several rounds of adjustment, we gradually made our queries more precise.

As for the presentation and writing process, each came with different challenges. Getting the slides' content to accurately reflect the findings, whilst being easy to understand and streamlined took many revisions, drafts, and restructuring. Once that was done, preparing the group for oratory was tricky. We mocked up the presentation a few times and shared insights on the kinds of vocabulary we needed to use and how to transition into each other's segments. To undertake the writing we decided to use a combination of git, LaTEX and VScode to build a collaborative environment. This set of tools if free to use and has ample documentation. It also produces sharp, academic looking papers. This decision was costly in terms of time, as the learning curve for LaTEX is rather hight and requires debugging and formatting is not required in regular document editors. On a brighter note, it does allow for easy exportation of citations from Zotero and automatic referencing, which makes managing source a breeze.

Conclusions and Reflections:

Overall, this project was both challenging and instructive. We started with an ambitious idea. From building our own predictive model to eventually recognised that this was unrealistic under our constraints. Instead of seeing this as a failure, we treated it as a learning experience about scoping the research question appropriately and understanding the importance of available data and resources.

The final scoping review allowed us to gain a broad overview of how geographical data and computational tools are currently used to determine restoration priorities in mangrove ecosystems. We could identify common modelling approaches, data sources, and gaps in the existing literature. At the same time, the process improved our skills in literature searching, critical reading and group collaboration. This later being likely the most important skill of all.

An interesting outcome of this project is that it highlighted the strong interdisciplinary nature of mangrove restoration research, which combines ecology, remote sensing,

computer science, and decision science. Even though we did not develop our own predictive model, the project allowed us to learn from the diverse methods used in the literature, understand how different computational and geospatial approaches contribute to mangrove conservation, and propose how these methods could potentially be combined in future work to achieve more robust and informative restoration planning.